Vector Magnitude & Direction: Cartesian to Polar

Right Triangle Trigonometry & Vectors • Standard: CCSS.MATH.HSN.VM.A.1

BOOK STORY OPENER: A Stick in Alexandria: How Eratosthenes Measured the Earth with a Shadow & Crosswind Vectors

In 240 BC, Eratosthenes measured a 7.2° shadow angle in Alexandria at noon on the summer solstice while a well in Syene 800 km away had zero shadow—calculating Earth's circumference within 1% using pure geometry! With the 3-disc Unit Circle Volvelle and [->ϑ,r] polar conversion, students resolve right triangles and aircraft crosswind vectors in a single keystroke.

Companion Book: Thinking in the Stack (SC-32 Manual) — Ch. 4: Trigonometry, Vectors & Polar Coordinates (A Stick in Alexandria)

1. CONCRETE MANIPULATIVE SETUP

Manipulative Kit: Unit Circle Trigonometry Volvelle

Align the Unit Circle Volvelle cursor to horizontal run 4 and vertical rise 3 (forming a right triangle with hypotenuse 5).

2. STUDENT PREDICTION (ASK BEFORE PRESSING KEYS)

Run = 4, Rise = 3. Predict magnitude: r = sqrt($4^2 + 3^2$) = 5. Predict angle: arctan(3/4) ~ 36.87°.

3. CALCULATOR ACTION (SC-32 KEYCAPS & STACK TRACE)

Keystroke Flow: [4] [ENTER] [3] [↶] [▸ϑ,r] [x⇄y]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[4]	4	Enter horizontal component x = 4
[ENTER]	4	4 lifted to register Y
[3]	3	Enter vertical component y = 3 in X
[↶] [▸ϑ,r]	36.87	Rectangular to Polar! Screen displays angle ϑ = 36.87°
[x⇄y]	5.00	Swap registers: Y displays vector magnitude r = 5.00!

4. MATHEMATICAL EXPLANATION & RPN ADVANTAGE

The calculator's built-in polar conversion executes both the Pythagorean magnitude and arctangent angle in a single microcode step.

Core Investigation Problem:

A surveyor walks 60 m East and 80 m North. Find the straight-line displacement magnitude and heading.

2D Vector Addition: Resultant Forces

Vector Addition & Resultant Forces • Standard: CCSS.MATH.HSN.VM.A.3

1. FOLLOW-UP EXTENSION SETUP & PREDICTION

Setup: Use the Unit Circle Volvelle to track component vectors F1 = <3, 2> and F2 = <1, 5>. Prediction: Add components: x_net = 3 + 1 = 4; y_net = 2 + 5 = 7. Magnitude r = sqrt(4^2 + 7^2) = sqrt(65) ~ 8.06.

2. EXTENSION KEYSTROKE SEQUENCE & STACK TRACE

3 ENTER 1 + 2 ENTER 5 + ↶ ▸θ,r x⇄y

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
3	3	Horizontal component x1 = 3
ENTER	3	Lifted to Y
1	1	Horizontal component x2 = 1
+	4	Net horizontal force x_net = 4 in X
2	2	Vertical component y1 = 2 (4 lifts to Y)
ENTER	2	2 lifted to Y; 4 in Z
5	5	Vertical component y2 = 5 in X
+	7	Net vertical force y_net = 7 in X; x_net = 4 in Y
↶ ▸θ,r	60.26	Resultant angle ϑ = 60.26°
x⇄y	8.06	Resultant force magnitude r = 8.06!

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Evaluating trigonometric functions in Radian mode when angles are given in Degrees (or vice versa).

Hands-On Block Remediation: Have students check the DEG/RAD indicator on the SC-32 status bar and verify on the Unit Circle Volvelle that sin(30°) = 0.5000 before starting vector problems.

Misconception 2: Swapping x and y components when converting rectangular coordinates to polar via [->ϑ,r].

Hands-On Block Remediation: On the SC-32, place the vertical component (y / Rise) in Register Y and the horizontal component (x / Run) in Register X before pressing [->ϑ,r].

4. EXTENSION MATHEMATICAL INSIGHT

Vectors add component-wise. Stack registers permit component accumulation followed immediately by polar resolution.

Trigonometry, Vectors & Polar/Rectangular Coordinates

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: VECTOR DECOMPOSITION & POLAR CONVERSION RULES

- A 2D vector v = <x, y> has horizontal run x in Y and vertical rise y in X.
- Rectangular to Polar [f:] [->θ,r]: Converts (x, y) into angle θ (in X) and magnitude r (in Y).
- Trig functions [SIN], [COS], [TAN] operate on the angle in register X.

Key Sequence: 4 ENTER 3 ↶ ▸θ,r x⇄y

Problem 1: Find magnitude and angle for vector <12, 5>.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-32

Problem 2: Find the hypotenuse of a right triangle with legs 7 and 24.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-32

Problem 3: Calculate sin(30°) on the calculator.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-32

Problem 4: A vector has magnitude 10 at angle 45°. What are its x and y components?

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-32

Unit 09: Trigonometry, Vectors & Polar/Rectangular Coordinates — Answer Key

Complete solutions and SC-32 verification keystrokes for Grades 9–11

1. CORE & EXTENSION INVESTIGATION SOLUTIONS

Core Investigation: A surveyor walks 60 m East and 80 m North. Find the straight-line displacement magnitude and heading.

Core Solution: Displacement: 100 m, Heading: 53.13°. Keystrokes: [6][0] [ENTER] [8][0] [f:][->ϑ,r] [x swap y]

Core Check Keys: 6 0 ENTER 8 0 ↶ ▸θ,r $x \rightleftarrows y$

Extension Challenge: Two cables pull on a post: Vector A is <5, 2> and Vector B is <-2, 4>. Find the resultant vector magnitude.

Extension Solution: Magnitude: 6.71. Resultant: <3, 6>.

Ext. Check Keys: 5 ENTER 2 +/− + 2 ENTER 4 + ↶ ▸θ,r $x \rightleftarrows y$

2. STUDENT LAB WORKSHEET SOLUTIONS (PROBLEMS 1–4)

Problem 1: Find magnitude and angle for vector <12, 5>.

Expected Answer: ϑ = 22.62°, r = 13

SC-32 Verification Keys: 1 2 ENTER 5 ↶ ▸θ,r

Problem 2: Find the hypotenuse of a right triangle with legs 7 and 24.

Expected Answer: 25

SC-32 Verification Keys: 7 ENTER 2 4 ↶ ▸θ,r $x \rightleftarrows y$

Problem 3: Calculate sin(30°) on the calculator.

Expected Answer: 0.5

SC-32 Verification Keys: 3 0 SIN

Problem 4: A vector has magnitude 10 at angle 45°. What are its x and y components?

Expected Answer: x = 7.07, y = 7.07

SC-32 Verification Keys: 1 0 ENTER 4 5 ↷ ▸y,x

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.